STAT 527
More on AIC and BIC

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Recall that in the context of linear regression, we defined

$$AIC := \frac{RSS_p}{\hat{\sigma}^2} + 2p'; \tag{1}$$

$$BIC := \frac{RSS_p}{\hat{\sigma}^2} + (\ln n)p'; \tag{2}$$

AIC is also known as Mallow's C_p in the context of linear regression. We

shall explain how (1) and (2) can be extended to general parameter estimations problems, and provide justifications.

Setup

A general model identification problem, with unknown parameter θ_0 (β_0 in linear regression) and iid samples $Z_1, Z_2, \ldots, Z_n \sim f_{\theta_0}$ ($(X_1, Y_1), \ldots, (X_n, Y_n)$ in linear regression.) Let $M \in \mathcal{M}$ denote a model (sparsity pattern in linear regression).

Algorithms

$\ln \hat{L} = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - (X\hat{\beta})_i)^2 + \text{const}.$

Denote by $\hat{L} := \prod_{i=1}^{n} f_{\hat{\theta}}(Z_i)$ the likelihood function, and k the dimension of the model M in which $\hat{\theta}$ lies. Then depending on the choice of criterion, we minimize the following cost functions:

$$AIC := -2 \ln \hat{L} + 2k; \tag{3}$$

$$BIC := -2 \ln \hat{L} + (\ln n)k. \tag{4}$$

Justification of AIC

We follow the original paper

Akaike 1974, "A new look at the statistical model identification."

Definition.

$$I(\theta_1, \theta_2) := D(f_{\theta_1} \| f_{\theta_2}) = \int \left(\ln \frac{f_{\theta_1}(z)}{f_{\theta_2}(z)} \right) f_{\theta_1}(z) dz. \tag{5}$$

Theorem. *For $\Delta\theta$ small, we have $I(\theta_0, \theta_0 + \Delta\theta) \approx \frac{1}{2}\|\Delta\theta\|_J^2 := \frac{1}{2}\Delta\theta^\top J \Delta\theta$ where the Fisher information matrix J is defined as*

$$J_{ij} = \mathbb{E}_{\theta_0} \left[\left. \frac{\partial \ln f_\theta(Z)}{\partial \theta_i} \right|_{\theta=\theta_0} \cdot \left. \frac{\partial \ln f_\theta(Z)}{\partial \theta_j} \right|_{\theta=\theta_0} \right] \tag{6}$$

$$= -\mathbb{E}_{\theta_0} \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \ln f_\theta(Z) \right]. \tag{7}$$

linear regression: $J = \left[\frac{\partial}{\partial \beta_i \partial \beta_j} \frac{(Y - X\beta)^\top}{2\sigma^2} \right]_{ij} = \frac{1}{\sigma^2} X^\top X$

Proof:

$$\frac{\partial}{\partial\theta_i} I(\theta_0, \theta_0+\Delta\theta) = \int -\frac{\frac{\partial}{\partial\theta_i} f_{\theta_0+\Delta\theta}(x)}{f_{\theta_0+\Delta\theta}(x)} \cdot f_{\theta_0}(x)\, dx$$

$$\frac{\partial^2}{\partial\Delta\theta_i \partial\Delta\theta_j} I(\theta_0, \theta_0+\Delta\theta)\Big|_{\Delta\theta=0} = \int -\frac{\frac{\partial^2}{\partial\Delta\theta_i \partial\Delta\theta_j} f_{\theta_0+\Delta\theta}(x)}{f_{\theta_0+\Delta\theta}(x)} \cdot f_{\theta_0}(x)\, dx \Big|_{\Delta\theta=0}$$

$$+ \int \frac{\frac{\partial}{\partial\Delta\theta_i} f_{\theta_0+\Delta\theta}(x) \frac{\partial}{\partial\Delta\theta_j} f_{\theta_0+\Delta\theta}(x)}{f^2_{\theta_0+\Delta\theta}(x)} \cdot f_{\theta_0}(x)\, dx \Big|_{\Delta\theta=0}$$

$$= 0 + \int \frac{\frac{\partial}{\partial\theta_i} f_\theta(x)\Big|_{\theta=\theta_0} \frac{\partial}{\partial\theta_j} f_\theta(x)\Big|_{\theta=\theta_0}}{f^2_{\theta_0}(x)} f_{\theta_0}(x)\, dx$$

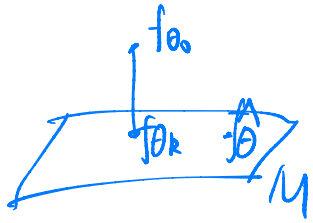
$$= J_{ij}$$

Theorem. *AIC approximately minimizes* $I(\theta_0, \hat{\theta})$.

In other words, approximately $I(\theta_0, \hat{\theta})$ differ from (3) only by an additive constant independent of $\hat{\theta}$. We prove the theorem in two steps:

Step 1: Suppose that M is a k-dimensional manifold in the total parameter space, and θ_k is the best approximator of θ_0 in M, that is,

$$\theta_k := \arg\min_{\theta \in M} I(\theta_0, \theta). \tag{8}$$



Let $\hat{\theta}$ be the MLE restricted to M. We have

$$\begin{aligned}
\mathbb{E}[2nI(\theta_0, \hat{\theta})] &\approx n\mathbb{E}[\|\theta_0 - \hat{\theta}\|_J^2] && (9)\\
&= n\|\theta_0 - \theta_k\|_J^2 + n\mathbb{E}[\|\theta_k - \hat{\theta}\|_J^2] && (10)\\
&\approx n\|\theta_0 - \theta_k\|_J^2 + k && (11)
\end{aligned}$$

$\mathbb{E}(\hat{\theta}) = \theta_k$

bias$^2/\sigma^2$ variance$/\sigma^2$

where (10) is the Pythagorean theorem, and (11) is a bias-variance decomposition. [Exercise: can you explicitly verify (11) in the concrete example of least squares estimation?]

Step 2: Returning to (11), we claim that $n\|\theta_0 - \theta_k\|_J^2$ is approximated by

$$2\Big(\sum_{i=1}^{n} \ln f_{\theta_0}(Z_i) - \sum_{i=1}^{n} \ln f_{\hat{\theta}}(Z_i)\Big) + k \tag{12}$$

which, together with (11), shows that $2nI(\theta_0, \hat{\theta})$ is approximately

$$n\|\theta_0 - \theta_k\|_J^2 + k \approx 2\Big(\sum_{i=1}^{n} \ln f_{\theta_0}(Z_i) - \sum_{i=1}^{n} \ln f_{\hat{\theta}}(Z_i)\Big) + k + k \tag{13}$$

$$= 2\sum_{i=1}^{n} \ln f_{\theta_0}(Z_i) + AIC. \tag{14}$$

Note that $2\sum_{i=1}^{n} \ln f_{\theta_0}(Z_i)$ does not depend on $\hat{\theta}$, so minimizing (14) is equivalent to minimizing AIC.

It remains to verify that (12) is approximately $n\|\theta_0 - \theta_k\|_J^2 \approx 2nI(\theta_0, \hat{\theta})$ in the case of linear regression: $(12) = -\frac{1}{\sigma^2}\|y - X\beta\|^2 + \frac{1}{\sigma^2}\|y - X\hat{\beta}\|^2 + k$

$$\approx -n + \frac{RSS}{\sigma^2} + k$$

$$\approx -n + \frac{1}{\sigma^2}(\text{bias}^2 + (n-k)\sigma^2) + k$$

$$= \frac{\text{bias}^2}{\sigma^2}$$

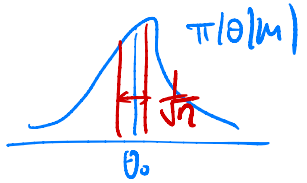
Justification of BIC

$P(x^n|M) = \int P(x^n|\theta)\pi(\theta|M)\,d\theta$
$= \int P(x^n|\theta, M)\,\pi(\theta|M)\,d\theta$

Theorem. $P(X^n|M) \approx \exp(-BIC/2+O(1))$ *for any smooth prior* $\pi(\theta|M)$, *as* $n \to \infty$.

Proof: Suppose that $\hat{\theta}$ is the MLE under model M. With high probability, as $n \to \infty$ we have

$$\|\hat{\theta} - \theta_0\| = O(\frac{1}{\sqrt{n}}). \tag{15}$$



Then by the Taylor expansion we have (note that $\hat{\theta}$ is MLE so the first derivative vanishes)

$$\ln p(X^n|\theta, M) \approx \ln \hat{L}_{\hat{\theta}}(X^n) - \frac{n}{2}\|\theta - \hat{\theta}\|^2_{J(\hat{\theta})} \tag{16}$$

for θ close to $\hat{\theta}$ ($\|\theta - \hat{\theta}\| = o(1)$). Moreover for θ close to $\hat{\theta}$ (and hence to θ_0)

the prior density $\pi(\theta|M)$ is approximately a constant, therefore

$$p(X^n|M) = \int p(X^n|M,\theta)\pi(\theta|M)d\theta \tag{17}$$

$$\approx \int p(X^n|M,\theta)\cdot const.\cdot d\theta. \tag{18}$$

The last last is a Gaussian integral in θ in view of (16), and hence

$$p(X^n|M) \approx \exp\left(\ln\hat{L} + \frac{k}{2}\ln 2\pi - \frac{1}{2}\ln|J(\hat{\theta})| - \frac{k}{2}\ln n + const.\right). \tag{19}$$

Note that $\ln\hat{L} - \frac{k}{2}\ln n = -\frac{1}{2}BIC$, and the other terms in (19) are not growing with n (the $O(1)$ term).

Terms that grow with n (BIC)

proof of (19):

$$\int p(x^n|M) = \int \exp\left(\ln \hat{L} - \frac{n}{2}\|\theta-\hat{\theta}\|_{J(\hat{\theta})} + O(1)\right) d\theta$$

$$\overset{(18)}{=} \int (2\pi)^{\frac{k}{2}} \det{}^{-\frac{1}{2}}(nJ) \exp\left(-\frac{n}{2}\|\theta-\hat{\theta}\|_{J(\hat{\theta})}\right) d\theta \cdot (2\pi)^{\frac{k}{2}} \det{}^{-\frac{1}{2}}(nJ) \hat{L}$$

$$\overset{\text{Gaussian Integral}}{=} \exp\left(\ln \hat{L} + \frac{k}{2}\ln 2\pi - \frac{1}{2}\ln\det(J) - \frac{k}{2}\ln n + O(1)\right)$$

$\ln \hat{L} - \frac{k}{2}\ln n$ → BIC

$\frac{k}{2}\ln 2\pi - \frac{1}{2}\ln\det(J) + O(1)$ → $O(1)$ terms (do not grow with n)